The Time Attraction Law (TAL), Release V20: A Conditional Design Prediction for a Coil Field-Energy Test with Twin Quartz Oscillators

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Abstract

The Time Attraction Law (TAL) postulates that the local rate of proper time depends on the top-hat averaged mass(-energy) density $\bar{\rho}_L$ through $f_L = \exp(-\gamma_L \bar{\rho}_L)$ with $\gamma_L = GL^2/c^2$, so that two clocks in environments differing by $\Delta\bar{\rho}_L$ acquire a fractional frequency difference $\Delta f/f \approx \gamma_L \Delta\bar{\rho}_L$ in magnitude. An extended form of the law includes the mass equivalent $B^2/(2\mu_0 c^2)$ of magnetic field energy. This paper converts that statement into a quantitative, fully documented design prediction for one fixed laboratory configuration specified by the author: eight vertical 2000-turn copper coils driven at 14 mA each, a main quartz oscillator (OCXO 8663 type) inside a granite shell and a twin reference oscillator 1.60 m away. The finite-solenoid field (current-sheet model) is $11.632900\,\mu$T at the main clock and $0.430539\,\mu$T at the reference. The differential field energy inside spheres of radius L is obtained with Gauss–Legendre radial panels, shared Fibonacci angular points and a direct $r = L/t$ substitution for the outer domain ($L \geq 2.6$ m). The largest calculated differential TAL signal of the isolated, polarity-even coil-B^2 term is $\Delta f/f \approx -1.20 \times 10^{-48}$ (-1.2017×10^{-48} at $L = 0.630957$ m on the archived grid); an independent Sobol volume integral gives $-1.2008435 \times 10^{-48}$ ($-0.0722\,\%$). Convergence, continuity and outer-integral checks, including an independent third-round review, are reported together with their remaining limitations. The result is a calculated order of magnitude within an idealised model, *not* a measured instrument sensitivity and *not* a TAL detection: no ON/OFF/$\pm I$ phase time series exists yet. A supplementary analysis of SPARC rotation curves contained in the repository (123-galaxy locked blind subsample) shows that the TAL factor at the published scales is numerically identical to the pure-baryon baseline.

Keywords: time attraction law; density-dependent proper time; top-hat averaged density; magnetic field energy; finite solenoid; quartz oscillator comparison; numerical quadrature; SPARC rotation curves

1 Introduction

In general relativity the rate of an ideal clock is set by the metric, and in the weak-field limit by the Newtonian potential, $d\tau \approx dt\,(1 + \Phi/c^2)$ [1]. Optical clocks now resolve this effect over laboratory height differences of tens of centimetres [2] and even across a millimetre-scale atomic

sample [3, 4], and clock comparisons are routinely used to search for physics beyond the standard model [5, 6]. A separate line of work has investigated scalar fields whose effective behaviour depends on the local matter density, such as chameleon models [7, 8].

The Time Attraction Law (TAL) [9, 10] is a phenomenological proposal of this broad type: it postulates that the local proper-time rate is an explicit function of the density averaged over a sphere of radius L. In its extended form, the effective density includes the mass equivalent of magnetic field energy [10]. The law therefore makes, in principle, a laboratory prediction for any configuration in which the magnetic energy around one clock can be switched while a second clock serves as reference.

The purpose of this paper is narrow and deliberately conservative. For one fixed apparatus specified by the author, it (i) states the TAL relations used, with exact equation and page references to the source documents; (ii) computes the differential TAL frequency shift produced by the coil field energy; (iii) documents the numerical method and its verification, including an independent third-round review [11]; and (iv) summarises, without extension, the SPARC rotation-curve comparison that exists in the project repository [12–14]. All numbers are taken from the archived calculation package (Zenodo V20, [11]; GitHub V18, [13]); file names are given so that each value can be traced.

The central result is a *conditional design prediction*: a calculated order of magnitude within an ideal current-sheet model. It is not a measurement, not an instrument sensitivity and not evidence for or against TAL. No clock phase data recorded under switched coil states exist at the time of writing.

2 Theoretical framework

2.1 Definitions

The following definitions are quoted from the TAL formalism document `Time_Attraction.pdf` [9] (copy in the repository root [13]). The top-hat averaged density at a point $\mathbf{x}$ is (Def. 1, Eq. (1), p. 3)

$$\bar{\rho}_L(\mathbf{x}) = \frac{3}{4\pi L^3} \int_{|\mathbf{x}'-\mathbf{x}|\le L} \rho(\mathbf{x}')\, \mathrm{d}^3x', \tag{1}$$

with $\bar{\rho}_L \to \rho$ for $L \to 0$. The coupling constant is (Def. 2, Eq. (2), p. 3)

$$\gamma_L = \frac{G_N L^2}{c^2} \qquad [\gamma_L] = \mathrm{m}^3\,\mathrm{kg}^{-1}, \tag{2}$$

and the time factor is (Def. 3, Eq. (7), p. 4)

$$f_L(\bar{\rho}_L) = \exp(-\gamma_L \bar{\rho}_L), \qquad \mathrm{d}\tau_L = f_L\,\mathrm{d}t, \qquad f_L \in (0,1]. \tag{3}$$

The source gives the weak-field clock prediction $\Delta f/f = \gamma_L \Delta\bar{\rho}_L$ (Eq. (13), p. 5) and an upper bound $\gamma < 10^{-22}\,\mathrm{m}^3\,\mathrm{kg}^{-1}$, equivalently $L < 367\,\mathrm{m}$, obtained from $\Delta f/f < 10^{-18}$ with $\Delta\rho = 10^4\,\mathrm{kg\,m}^{-3}$ (Eq. (3), p. 3). The same bound appears as $\gamma_{\mathrm{lab}} < 1.0 \times 10^{-22}\,\mathrm{m}^3\,\mathrm{kg}^{-1}$ in `Idovonzas_x5f_publikacio.pdf` (Eq. (19), p. 3) [10]. This bound is the author's own estimate from a generic clock-stability argument; it is not a constraint reported by a dedicated density-modulation experiment.

2.2 Field-energy term

The extended model of `Idovonzas_x5f_publikacio.pdf` (Sect. 7, Eqs. (15)–(16), p. 2) [10] adds the mass equivalent of energy densities to the effective density,

$$\rho_{\mathrm{tot}} = \rho_b + \rho_{\mathrm{DM}} + \frac{B^2}{2\mu_0 c^2} + \ldots, \qquad f(\mathbf{r},t) = \exp[-\gamma_L \rho_{\mathrm{tot}}(\mathbf{r},t)], \tag{4}$$

where the omitted term is the curvature contribution given in the source. Only the magnetic term $\rho_B = B^2/(2\mu_0 c^2)$ is varied in the configuration studied here.

2.3 Differential prediction for two clocks

For a static coil current I, the field energy enclosed by a sphere of radius L centred on a clock at $\mathbf{x}$ is

$$W(<L;\mathbf{x}) = \int_{|\mathbf{x}'-\mathbf{x}|<L} \frac{|\mathbf{B}(\mathbf{x}')|^2}{2\mu_0}\,\mathrm{d}^3x', \tag{5}$$

so that, by Eq. (1), $\bar{\rho}_{B,L}(\mathbf{x}) = W(<L;\mathbf{x})/(c^2 V_L)$ with $V_L = 4\pi L^3/3$. With $\Delta W(L) = W(<L;\mathbf{x}_{\mathrm{main}}) - W(<L;\mathbf{x}_{\mathrm{ref}})$ the signed differential prediction used throughout is

$$\left(\frac{\Delta f}{f}\right)_{\mathrm{diff}} = \mathrm{expm1}[-\gamma \Delta\bar{\rho}_{B,L}] = \mathrm{expm1}\left[-\gamma \frac{\Delta W(L)}{c^2 V_L}\right] \approx -\frac{3G_N}{4\pi c^4}\frac{\Delta W(L)}{L} \quad (\gamma = \gamma_L), \tag{6}$$

where the last form follows algebraically from Eq. (2). A negative value means that the main clock runs slower than the reference. Because $\mathbf{B} \propto I$, $\Delta W \propto I^2$; the package reports $\Delta W(L)/I^2$ in $\mathrm{J\,A^{-2}}$. Two readings of the coupling are evaluated: (a) $\gamma = \gamma_L = G_N L^2/c^2$, with L scanned over 81 logarithmically spaced values between 0.01 and 100 m, and (b) a fixed $\gamma = 10^{-22}\,\mathrm{m^3\,kg^{-1}}$ as a cross-check at the clock bound. For each reading the largest $|\Delta f/f|$ over the L grid is quoted. The grid maximum is not claimed to be an accurately determined continuous maximum location.

2.4 Polarity reversal and background field

The pure coil-B^2 term is even under polarity reversal. In a constant background field, the change in energy density also contains a $2\mathbf{B}_{\mathrm{bg}}\cdot\mathbf{B}_{\mathrm{coil}}$ cross term; ideally, averaging equal-magnitude $\pm I$ states relative to OFF cancels this term. The reported -1.20×10^{-48} value refers to this isolated even term, not to the full prediction for every possible simple ON/OFF measurement. Explicitly, for an unchanged background field $\mathbf{B}_{\mathrm{bg}}$ and coil field $\pm\mathbf{b}$,

$$\tfrac{1}{2}\left\{\left[|\mathbf{B}_{\mathrm{bg}}+\mathbf{b}|^2 - |\mathbf{B}_{\mathrm{bg}}|^2\right] + \left[|\mathbf{B}_{\mathrm{bg}}-\mathbf{b}|^2 - |\mathbf{B}_{\mathrm{bg}}|^2\right]\right\} = |\mathbf{b}|^2, \tag{7}$$

within the linear frequency-shift approximation [11]. The calculation does not establish an actual background-field measurement, $\pm I$ symmetry or a complete prediction for a simple ON/OFF experiment; those require experimental data.

3 Methodology

3.1 Experimental geometry

All geometry and electrical inputs are author-supplied values recorded in `report_hu.md` (Sect. 1, 1.1, 1.2) and hard-coded in `code/final_fixed.py` of the V20 package [11]. They

Table 1: Fixed configuration (author-supplied inputs unless marked "derived"). Source: `report_hu.md` Sect. 1; `code/final_fixed.py`; `csv/fixed_case_numbers.json`.

Item	Value
Coils	8, vertical axes, parallel connection, identical current direction; axis circle radius 0.45 m (coil-body inner edge at 0.30 m, outer edge at 0.60 m)
Single coil	length 400 mm; 2000 turns; 0.15 mm bare Cu wire; plastic former; flange (body) outer diameter 300 mm; measured winding diameter 291.7 mm (Appendix A)
Resistance, current	$R = 1789\,\Omega$ per coil (all eight identical); $I = 14$ mA per coil
Supply	24 V, 30 Ah battery; U and I logged continuously for 12 h (author)
Derived electrical	$14\,\text{mA} \times 1789\,\Omega = 25.05$ V; $24\,\text{V}/1789\,\Omega = 13.4$ mA; 0.351 W per coil, 2.81 W total; source current 0.112 A
Derived wire data	wire length $N\pi D = 1832.8$ m per coil; R_{20} from geometry $1788.160\,\Omega$ ($-0.0470\,\%$ vs. $1789\,\Omega$); Cu mass (8 coils) 2.303 kg
Main clock	OCXO, 10 MHz, SC-cut 3rd overtone; centred, 300 mm above the bottom plane of the coils; housing 58–62 °C (author)
Granite shell	uniform cylindrical shell, $r = 0.09$–0.22 m, $h = 0.50$ m, open, centred on the clock; 2700 kg m^{-3}; derived mass 170.9 kg; 8 cm gap to the coil bodies
Faraday capsule	cylindrical; inner ∅ 10 cm, outer ∅ 16 cm, height 24 cm; layers (outside to inside): 10-layer Al foil, 20 m of 1.5 mm bare Cu wire, Al foil, Cu coating, expanded-polystyrene (Hungarocell) sleeve, 1.2 mm cast Al housing; 1 cm gap to the granite
Reference clock	full twin OCXO in an identical capsule, same height and temperature, no granite; 1 m from the outer coil-body surface, i.e. 1.60 m from the main clock along one coil axis direction
Signal logging	PicoScope 2205A and Rigol DS1202ZE oscilloscopes (author)

are summarised in Table 1.

The oscillators are specified by the author as OCXO 8663XS (10 MHz, 12 V, SC-cut, 3rd overtone). A Micro Crystal 8663XS datasheet could not be located; the package therefore compares the author's parameters with the Oscilloquartz OCXO 8662/8663 technical specification (Edition March 2003) [15]: short-term stability $< 1 \times 10^{-11}$ (0.2–10 s), ageing 2×10^{-10}/day, temperature stability $< 4 \times 10^{-9}$ p-p, operating range -20 to $+70$ °C. The datasheet gives no magnetic sensitivity. Whether the operating range refers to the housing temperature is not stated in the datasheet, so the comparison with the measured 58–62 °C is conditional.

3.2 Finite-solenoid field

Each coil is modelled as an ideal cylindrical current sheet of radius $a = D/2 = 0.14585$ m, length $\ell = 0.400$ m and NI ampere-turns. The off-axis field uses the Derby–Olbert closed form in terms of Bulirsch's generalised complete elliptic integral [16] (`code/fields.py`, 12 iterations of the `cel` recursion). The coil centres lie on a circle of radius 0.45 m, 0.1 m below the clock plane, so that the clock is 300 mm above the bottom plane of the coils. The field is the vector sum of the eight coils. As an independent check, `V18/make_revision3_checks.py` sums

8×2000 discrete circular loops (complete elliptic integrals K, E from SciPy [17]); the reviewer additionally compared the package field at five points with a separate circular-loop integral [11].

3.3 Top-hat finite-source integration

`code/tophat_finite_source.py` implements Eqs. (1)–(3) for finite sources (constants $G = 6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$, $c = 299\,792\,458\,\mathrm{m\,s^{-1}}$ as in [9]), including the ball volume V_L and the exact ratio $\mathrm{expm1}(-\gamma\Delta\bar{\rho}_L)$. For static masses (copper windings, granite) the enclosed mass fraction of a sphere of radius L around each clock is estimated with shared Monte Carlo samples of the source volume (400 000 samples for the thin copper shells, 10^6 for the granite), and $\Delta\bar{\rho}_L = M\,(\phi_{\mathrm{main}} - \phi_{\mathrm{ref}})/V_L$ is formed for the same 81 L values. The capsules are identical for both clocks, so their differential static term is zero by construction.

3.4 Differential field-energy integrator

`code/energy_integrator.py` computes $\Delta W(L)$ for $I = 1$ A without radial interpolation. The shell function $S(r) = 4\pi r^2 \langle |\mathbf{B}|^2 \rangle_{S^2}/(2\mu_0)$ is evaluated at each radius by averaging over a Fibonacci spherical point set [18]. The *same* angular points are used for both centres at the same r (shared angular points): $n_{\mathrm{hi}} = 24\,000$ points on shells with $0.25 \le r \le 2.4$ m, which intersect the coils, and $n_{\mathrm{lo}} = 6000$ points elsewhere. Radial integration uses 8-point Gauss–Legendre panels with breakpoints at 0, 0.25, 0.75, 2.6 and 3.0 m (12, 50, 92 and 8 panels per interval in the base setting). Three regimes are used:

1. $L \le 0.95$ m (inner form; the reference sphere contains no coil): $\Delta W = W_{\mathrm{main}}(<L) - W_{\mathrm{ref}}(<L)$.
2. $0.95 < L < 2.6$ m (route A, outer difference): $\Delta W = W_{\mathrm{ref}}(>L) - W_{\mathrm{main}}(>L)$, with panels from L to 3 m followed by a tail $\int_3^\infty$ evaluated with $r = 3/t$.
3. $L \ge 2.6$ m (route B): direct outer difference with the substitution $r = L/t$, $t \in (0, 1]$, 32-point Gauss–Legendre in t ($n_t = 32$), with no truncation and no interpolation.

The outer forms use no separately estimated total energy, so the difference of two independently estimated total energies no longer enters the outer branch (the earlier $0.01268\,\mathrm{J\,A^{-2}}$ offset near $L \approx 3$ m is thereby removed [11]).

3.5 Convergence and independent checks

Four variants were run: $n_{\mathrm{hi}} = 24\,000$ (base), 6000 and 12 000, and 12 000 with doubled radial panel density; $n_{\mathrm{lo}} = 6000$ and $n_t = 32$ are identical in all four. The spread is reported as the sample standard deviation $s = \sqrt{\sum (x_i - \bar{x})^2/(n-1)}$ with $n = 4$. Continuity of routes A and B was tested at $L = 2.6$–3.01 m. As an independent cross-check, a shared-sample volumetric Monte Carlo estimate (1.5×10^6 points per L; difference of the two ball indicator functions weighted by the field energy density) was computed at $L = 0.316$, 0.631, 1.0 and 2.0 m (`csv/convergence_mc_shared_samples.csv`). In the third-round review [11], the main peak was additionally recomputed with a Cartesian Sobol cubature [19] (4×2^{19} points for the main sphere, 4×2^{12} points for the smooth reference-sphere integral), and the far-field energy difference with separate Gauss–Legendre polar and radial quadrature and uniformly spaced azimuthal angles.

3.6 Clock-comparison reference quantities

The author observed that the Lissajous figure of the two OCXOs completes one 360° cycle in 1 h 5 min (3900 s). Provided this is a true 2π electrical phase cycle, the conversion is $\Delta f = 1/3900\,\mathrm{Hz} = 0.256410\,\mathrm{mHz}$ and $|\Delta f/f| = 2.56410 \times 10^{-11}$ at 10 MHz. This is a *mean beat offset, not a noise floor*. For orientation only, the package also compares the signal with the cited systematic uncertainty of a Sr lattice clock, 8.1×10^{-19} [20], with the Sr two-clock stability 4.8×10^{-17} at 1 s [21], and evaluates the Sr quadratic Zeeman shift with the coefficient $-23.0\,\mathrm{MHz\,T^{-2}}$ [22]. The Sr value is a cited systematic uncertainty, not a universal noise floor. Finally, "main-clock-equivalent" coefficients are defined as the linear and quadratic magnetic coefficients $k_1 = |\Delta f/f|_{\mathrm{offset}}/B_{\mathrm{main}}$ and $k_2 = |\Delta f/f|_{\mathrm{offset}}/B^2_{\mathrm{main}}$ that would ideally produce a frequency change equal in magnitude to the mean beat offset; they are not measured coefficients and not upper bounds.

3.7 SPARC rotation-curve data

The repository contains a supplementary analysis (V19, [12]; `tal_factor_7th_model.md` and `hetedik_modell_kimenet/` in [13]) that applies the TAL factor as a seventh model inside the hash-locked SPARC baseline runner of the repository's `v17_package/github/` directory. The rotation curves are SPARC `Rotmod_LTG.zip` (175 disk galaxies) [14], SHA-256 `0a80cc90...f588`; the runner (`sparc_full_runner.py`) uses a locked blind subsample `blind_123.txt` of 123 galaxies with 2596 radii. The error model is $\sigma = \max(0.05\,V_{\mathrm{obs}}, 20\,\mathrm{km\,s^{-1}})$, $\chi^2 = \sum[(V_{\mathrm{pred}} - V_{\mathrm{obs}})/\sigma]^2$, $\chi^2_{\mathrm{red}} = \chi^2/(N-1)$. The primary TAL model is $V^2_{\mathrm{pred}} = f_L(\bar{\rho})\,V^2_{\mathrm{bar}}$ with $L_0 = 1.2 \times 10^{-11}$ m, where $\bar{\rho}$ is a disk-plus-bulge midplane density proxy ($h_z = 0.2$ kpc, $\Upsilon_{\mathrm{disk}} = 0.5$, $\Upsilon_{\mathrm{bulge}} = 0.7$ inside $\bar{\rho}$ only). How f_L enters $V(R)$ is an implementation choice not specified in the published material; an inverse coupling $V^2 = V^2_{\mathrm{bar}}/f_L$ was run as a sensitivity check. Per-galaxy χ^2_{red} values were compared with two-sided Wilcoxon signed-rank and exact sign tests.

4 Results

4.1 Magnetic field

Table 2 lists the field values. The package current-sheet field and the independent discrete-loop sum agree to -4.3×10^{-8} (main) and -7.6×10^{-9} (reference) relative (`V18/revision3_checks.json`); in the reviewer's separate circular-loop test the largest relative discrepancy over five points was 1.04×10^{-14} [11].

4.2 Differential TAL signal

Figure 1 shows the signed differential prediction, Eq. (6), as a function of L. $\Delta W(L)$ is positive at all evaluated L (no negative large-L tail occurs). With $\gamma = \gamma_L$ the largest magnitude on the grid is

$$\left(\frac{\Delta f}{f}\right)_{\mathrm{diff}} = -1.2017 \times 10^{-48} \quad \text{at } L = 0.630957\ \mathrm{m}, \tag{8}$$

i.e. $\approx -1.20 \times 10^{-48}$, corresponding to $k = -6.131177672606294 \times 10^{-45}\,\mathrm{A^{-2}}$ and $\Delta W/I^2 = 1.961\,\mathrm{J\,A^{-2}}$ at that L (`csv/fixed_case_numbers.json`, `TAL_diff`;

Table 2: Magnetic field of the eight coils at 14 mA per coil (current-sheet model, $D = 291.7$ mm). Sources: `csv/fixed_case_numbers.json` (`Bpa`, `Bref_pa`, `rows`), `csv/fixed_case_and_feasibility.csv`, `revision3_review_EN.md` §2.

Quantity	Value
B at main clock	$11.632900\,\mu$T ($0.8309\,$mT A^{-1}, vertical)
B at reference clock (1.60 m)	$0.430539\,\mu$T ($30.75\,\mu$T A^{-1})
B inside a coil	$71.1\,\mu$T
Magnetic pressure in a coil, $B^2/2\mu_0$	2.0×10^{-3} Pa

Table 3: Selected rows of `csv/tal_differential_vs_L.csv` (base run; values as archived, 4 significant digits). $\Delta\bar{\rho}_L$ is main minus reference.

L (m)	$\Delta W/I^2$ (J A^{-2})	$\Delta\bar{\rho}_L/I^2$ (kg m^{-3} A^{-2})	$\Delta f/f$, γ_L, 14 mA	$\Delta f/f$, $\gamma = 10^{-22}$, 14 mA
0.1	1.184×10^{-3}	3.145×10^{-18}	-4.577×10^{-51}	-6.164×10^{-44}
0.3162	6.382×10^{-2}	5.361×10^{-18}	-7.803×10^{-50}	-1.051×10^{-43}
0.5623	1.582	2.363×10^{-17}	-1.087×10^{-48}	-4.631×10^{-43}
0.631	1.961	2.074×10^{-17}	-1.202×10^{-48}	-4.065×10^{-43}
1	2.177	5.782×10^{-18}	-8.416×10^{-49}	-1.133×10^{-43}
1.995	0.4398	1.471×10^{-19}	-8.522×10^{-50}	-2.883×10^{-45}
3.162	1.620×10^{-3}	1.361×10^{-22}	-1.981×10^{-52}	-2.667×10^{-48}
10	2.778×10^{-6}	7.378×10^{-27}	-1.074×10^{-55}	-1.446×10^{-52}
100	2.636×10^{-11}	7.003×10^{-35}	-1.019×10^{-61}	-1.373×10^{-60}

`csv/tal_differential_vs_L.csv`). Accumulated over one day this is -1.04×10^{-43} s (-5.19×10^{-44} s per 12 h). With the fixed $\gamma = 10^{-22}\,$m^3 kg^{-1} cross-check the maximum is -4.63×10^{-43} at $L = 0.5623$ m (-4.00×10^{-38} s/day). Selected values are listed in Table 3.

4.3 Independent check of the main peak

A targeted rerun of the archived integrator at $L = 0.630957$ m reproduces -1.2017108×10^{-48}, with a relative difference of 2.2×10^{-16} from the archived value (review [11]; the regenerated `V18/revision3_checks.json` gives $-1.2017108238308338\times10^{-48}$ against the JSON value $-1.2017108238308337\times10^{-48}$). The reviewer's independent volumetric Sobol integral at the same L gives -1.2008435×10^{-48}, a $-0.0722\,\%$ difference in signal magnitude, with a standard error of 3.04×10^{-52} estimated from Sobol scrambles. This standard error is not the uncertainty of the full model: it does not include uncertainties in coil geometry, current, the actual winding or the environment.

4.4 Convergence

Table 4 reproduces `csv/convergence_dW.csv`; Fig. 2 shows the four variants together with the shared-sample Monte Carlo values of Table 5. At the peak ($L = 0.631$ m) the four-run relative spread is 1.05×10^{-4}; the largest relative spread among the tabulated points, 6.55×10^{-3}, occurs at $L = 2$ m, inside the region where the reference-centred shells intersect the coils.

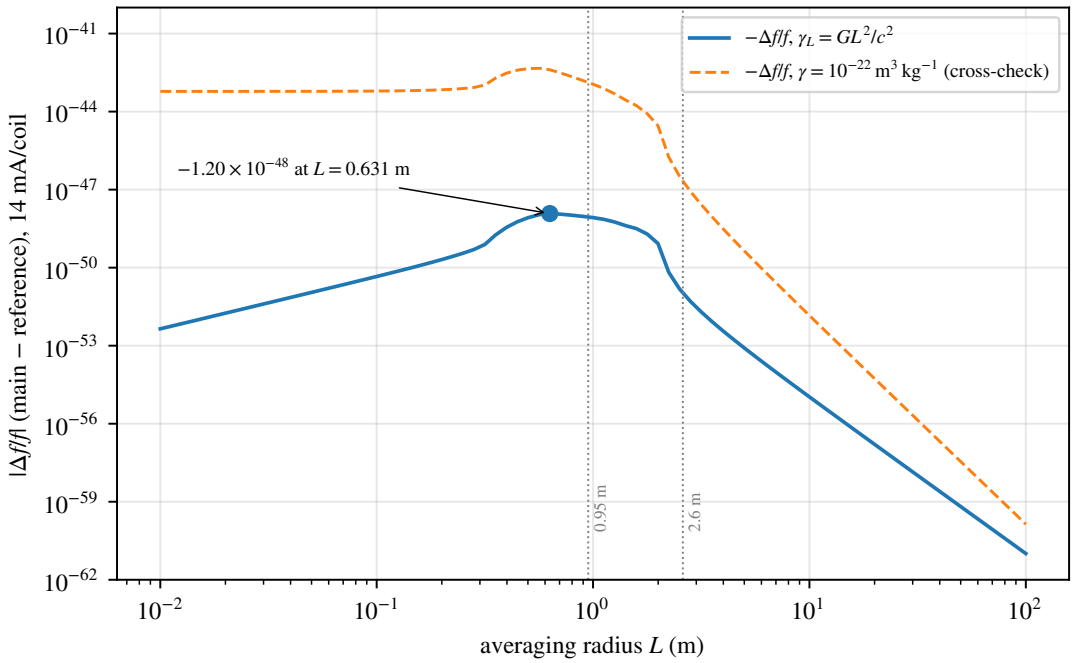


Figure 1: Magnitude of the signed differential TAL prediction (main minus reference; all values negative) of the isolated polarity-even coil-B^2 term at 14 mA per coil, versus averaging radius L (81 grid points). Solid: $\gamma = \gamma_L = GL^2/c^2$; dashed: fixed $\gamma = 10^{-22}\,\mathrm{m^3\,kg^{-1}}$. Dotted vertical lines mark the inner/outer switch (0.95 m, where a small matching residual remains, Sect. 4.7) and the start of the direct $r = L/t$ outer integral (2.6 m). Regenerated from `csv/tal_differential_vs_L.csv`.

4.5 Continuity of the outer routes near 3 m

Routes A (panels plus $r = 3/t$ tail) and B ($r = L/t$) were evaluated on the overlap $L = 2.6$–3.01 m (Table 6). The relative differences at $L = 2.99$, 3.00 and 3.01 m are 0, -7.4×10^{-16} and -1.5×10^{-15}; on 2.6–2.9 m they are at most 2.9×10^{-15} in magnitude. The reviewer's targeted rerun found the two routes consistent to approximately 10^{-15} relative across 2.6–3.01 m [11].

4.6 Independent outer-integral comparison

Table 7 reproduces the reviewer's comparison of the archived integrator with a separate integral that uses Gauss–Legendre polar and radial quadrature with uniformly spaced azimuthal angles (not the archived Fibonacci average) and maps the infinite domain with $r = L/t$ [11]. The field function is shared with the archive. The coarse/fine grids of the separate integral differ by at most a few 10^{-6} relative at the nearby points; this is a convergence test, not a proof of an absolute error bound. At $L = 100$ m the residual decreases with angular refinement: 6000/12 000/24 000 far-field angular points give 2.6364994, 2.6357023 and $2.6355308 \times 10^{-11}\,\mathrm{J\,A^{-2}}$, with 32 and then 64 radial t points. No certified error bound is claimed for every L or for arbitrary distances.

4.7 Matching residual at $L = 0.95$ m

The inner/outer switch at $L = 0.95$ m is not exactly continuous. The archived inner and outer values are 2.17738947 and $2.17597607\,\mathrm{J\,A^{-2}}$; the matching residual is $0.00141340\,\mathrm{J\,A^{-2}}$, i.e. 0.0649 % of the inner value. It equals, to $8.9 \times 10^{-16}\,\mathrm{J\,A^{-2}}$, the difference between the

Table 4: Convergence of $\Delta W(L)/I^2$ over four integrator variants ($n_{\mathrm{hi}} = 24\,000$ base, 6000, 12 000, and 12 000 with doubled radial panels). s is the sample standard deviation ($n-1$, $n=4$). Source: `csv/convergence_dW.csv`.

L (m)	Base value (J A^{-2})	Range, min–max (J A^{-2})	s (J A^{-2})	s/mean
0.1	0.00118391	0.00118391–0.00118391	0[a]	0
0.3162	0.0637737	0.0637725–0.0637856	6.85×10^{-6}	1.07×10^{-4}
0.631	1.96126	1.96091–1.96135	2.05×10^{-4}	1.05×10^{-4}
1	2.17691	2.17526–2.18903	6.53×10^{-3}	3.00×10^{-3}
2	0.426287	0.426248–0.431987	2.80×10^{-3}	6.55×10^{-3}
2.6	0.00713721	identical in all 4 runs	0[b]	0[b]
3.162	0.00162106	identical in all 4 runs	0[b]	0[b]
10	2.77763×10^{-6}	identical in all 4 runs	0[b]	0[b]
31.62	8.37811×10^{-9}	identical in all 4 runs	0[b]	0[b]
100	2.6365×10^{-11}	identical in all 4 runs	0[b]	0[b]

[a] All four runs agree at $L = 0.1$ m; the varied n_{hi} acts only on shells with $0.25 \le r \le 2.4$ m.
[b] The four runs use identical nlo and nt settings in this region. The zero sample standard deviation is therefore not a far-field error estimate; far-field convergence must be tested by varying nlo and nt separately.

Table 5: Shared-sample volumetric Monte Carlo check (1.5×10^6 points per L). Source: `csv/convergence_mc_shared_samples.csv`.

L (m)	MC $\Delta W/I^2$ (J A^{-2})	MC 1σ (J A^{-2})	$\sigma/\Delta W$	GL integrator (J A^{-2})	(GL−MC)/σ
0.3162	0.06646	0.001444	2.17 %	0.06377	−1.86
0.631	1.949	0.0194	0.99 %	1.961	+0.61
1	2.154	0.02455	1.14 %	2.177	+0.92
2	0.4266	0.01722	4.04 %	0.4263	−0.02

two total-energy estimates, 2.237861 (main) and 2.236448 J A^{-2} (reference) (`csv/fixed_case_numbers.json`, `jump_LJ`, `W_total`; `V18/revision3_checks.json`, `L095_residual`), and is attributed to the angular quadrature error on reference-centred shells that intersect the coils; it affects only the 0.95–2.4 m range. This remains a numerical limitation: the curve is not continuous to machine precision, and machine-precision agreement applies only to the 2.6–3.01 m A/B overlap.

4.8 Comparison scales

Table 8 relates the predicted signal to the reference quantities of Sect. 3.6. All ratios are arithmetic comparisons; none is a detection threshold of the present apparatus.

The Sr quadratic Zeeman shift at 11.63 μT, -7.25×10^{-18}, is quoted as a cited systematic for an atomic clock in the same field; it is not a universal noise floor and does not apply to quartz, which has no atomic Zeeman shift. The quoted white-FM averaging time is an algebraic extrapolation of the chosen formula, not an experimental time prediction. The main-clock-equivalent coefficients only indicate what ideal magnetic frequency change would equal the observed mean beat offset; without a measured ON/OFF uncertainty and separation of

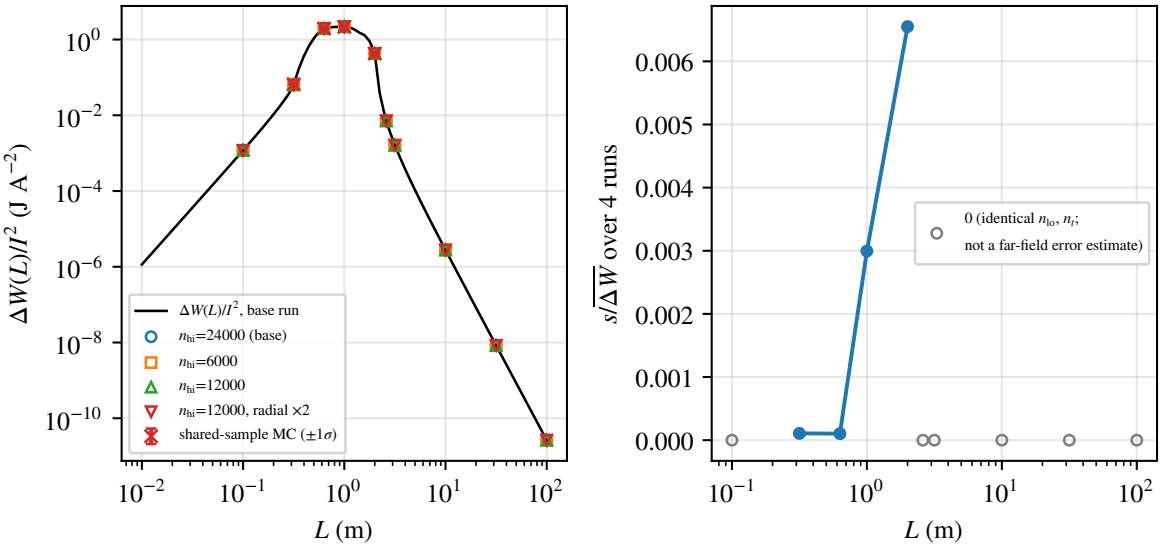


Figure 2: Left: $\Delta W(L)/I^2$ from the base run (line), the four integrator variants at the check radii (open symbols) and the shared-sample Monte Carlo estimates with 1σ error bars (crosses). Right: relative sample standard deviation over the four variants. Open grey circles are exact zeros obtained with identical n_{lo}, n_t settings and are not far-field error estimates. Regenerated from `csv/convergence_dW.csv`, `csv/convergence_mc_shared_samples.csv` and `csv/tal_differential_vs_L.csv`.

environmental couplings no upper bound follows from them.

4.9 Static mass terms

The granite shell and the copper windings produce static, not modulated, offsets (`csv/fixed_case_numbers.json`, `granite`, `cu_static`; `csv/granite_static_differential.csv`, `csv/copper_coils_static_differential.csv`). For the granite (170.9 kg), the general-relativistic potential term is $\Phi/c^2 = -6.28 \times 10^{-25}$ at the main clock and -7.92×10^{-26} at the reference (difference -5.49×10^{-25}); the TAL extreme is -9.6×10^{-26} (γ_L, $L = 0.28$ m) and -2.1×10^{-19} ($\gamma = 10^{-22}$, $L = 0.22$ m). For the copper (2.303 kg), the GR terms are -3.68×10^{-27} and -1.09×10^{-27} (difference -2.60×10^{-27}) and the TAL extremes -6.2×10^{-28} (γ_L) and -2.3×10^{-22} ($\gamma = 10^{-22}$). The capsules are identical, so their differential static term is zero. The mass and material of the plastic coil formers are unknown; their static differential term could not be calculated.

4.10 Higher-current cases

Table 9 reproduces the author's list of higher currents. None is feasible with the available supply, and all exceed the continuous-current limit (0.405 A) while remaining below the free-air Preece fusing current (4.65 A). The Onderdonk estimate is only an idealized adiabatic fusing time; no safe pulse duration follows from it, and the calculated high-current cases are not operating recommendations. Even in these cases the signal is $\leq 5.5 \times 10^{-44}$ in magnitude.

Table 6: Route A versus route B for the outer difference. Source: `csv/continuity_test.csv`.

L (m)	Route A (J A^{-2})	Route B (J A^{-2})	$(A-B)/B$
2.6	$7.137211082279\times10^{-3}$	$7.137211082279\times10^{-3}$	-2.917×10^{-15}
2.7	$5.182239219640\times10^{-3}$	$5.182239219640\times10^{-3}$	-2.008×10^{-15}
2.8	$3.881360582648\times10^{-3}$	$3.881360582648\times10^{-3}$	-1.006×10^{-15}
2.9	$2.978621592572\times10^{-3}$	$2.978621592572\times10^{-3}$	-4.368×10^{-16}
2.99	$2.387273260006\times10^{-3}$	$2.387273260006\times10^{-3}$	0
3.00	$2.331314942858\times10^{-3}$	$2.331314942858\times10^{-3}$	-7.441×10^{-16}
3.01	$2.277039189332\times10^{-3}$	$2.277039189332\times10^{-3}$	-1.524×10^{-15}

Table 7: Outer-integral comparison from the independent third-round review (`revision3_review_EN.md` §3, [11]).

L (m)	Archived integrator ΔW (J A^{-2})	Separate integral (J A^{-2})	Rel. discrepancy
2.6	0.00713721108	0.00713720352	+0.000106 %
2.818383	0.00369084579	0.00369083957	+0.000169 %
3.00	0.00233131494	0.00233130964	+0.000227 %
3.162	0.00162105726	0.00162105265	+0.000284 %
10	2.77763231×10^{-6}	2.77754429×10^{-6}	+0.00317 %
31.62	8.37811251×10^{-9}	8.37717664×10^{-9}	+0.0112 %
100	2.63649938×10^{-11}	2.63556282×10^{-11}	+0.0355 %

4.11 SPARC rotation curves (repository analysis)

The repository does *not* contain rotation-curve fit results for all 175 SPARC galaxies. The script `Repoduce SPARK 175 v16 1` in the repository root generates synthetic exponential profiles and mock "observed" velocities with random scatter when the SPARC directory is absent; it is not an analysis of SPARC data, and no results from it are used here. The only SPARC analysis with archived results is the seventh-model comparison on the 123-galaxy locked blind subsample (2596 radii) summarised in Table 10 (`hetedik_modell_kimenet/comparison.json`, `hetedik_modell_kimenet/run_meta.json`, `tal_factor_7th_model.md`; [12, 13]). The six baseline outputs were reported as byte-identical to `v17_package/github/run_output/`.

For the primary model, $\max\gamma_0\bar{\rho} = 6.3\times10^{-67}$ and $\max|f_L-1| = 0$ in double precision at all 2596 points; all predicted velocities and all 123 per-galaxy $\chi^2_{\rm red}$ values are identical to the `vbar` baseline (123 ties). It therefore ranks exactly where `vbar` ranks: better than vbar_095 (97 vs. 26 galaxies; Wilcoxon $p = 2.57\times10^{-4}$), no_gas (110 vs. 13) and disk_only (111 vs. 12), and worse than vbar_105 (32 vs. 91; Wilcoxon $p = 0.0146$). At $L = 2.36$ kpc, $|f_L-1| \le 2.33\times10^{-5}$. The fitted forward scale lowers the median but is worse than `vbar` in 81 of 123 galaxies (sign test $p = 5.6\times10^{-4}$) and was obtained in-sample. Order-one scales (tens of Mpc, or $\gamma = 10^{23}$) worsen the per-galaxy fits (`tal_factor_7th_model.md` §4.3). These results are consistent with the source's own statement that the galactic correction at laboratory-scale γ is of order 10^{-34} and negligible (`Time_Attraction.pdf`, Sect. 8, Eq. (14), p. 5 [9]).

Table 8: Comparison quantities for the fixed case (14 mA per coil). Source: `csv/fixed_case_and_feasibility.csv`, `csv/fixed_case_numbers.json`, `report_hu.md` §2.

Quantity	Value
Mean beat offset (65-min Lissajous cycle; *mean beat offset, not a noise floor*)	2.56410×10^{-11}
$\lvert\Delta f/f\rvert$ / mean beat offset: γ_L; $\gamma = 10^{-22}$	4.69×10^{-38}; 1.81×10^{-32}
$\lvert\Delta f/f\rvert$ / datasheet short-term stability 1×10^{-11} [15]	1.2×10^{-37}
$\lvert\Delta f/f\rvert$ / cited Sr systematic uncertainty 8.1×10^{-19} [20]	1.48×10^{-30}
Sr white-FM averaging time, $k = 2$, $\sigma(1\,\mathrm{s}) = 4.8 \times 10^{-17}$ [21] (algebraic extrapolation only)	1.6×10^{57} yr
Sr quadratic Zeeman shift at the main-clock field [22]	-7.25×10^{-18}
Field below which the Sr Zeeman shift stays under 8.1×10^{-19}	$< 3.9\,\mu$T
Main-clock-equivalent coefficient k_1 (linear)	$2.20 \times 10^{-6}\,\mathrm{T}^{-1}$
Main-clock-equivalent coefficient k_2 (quadratic)	$0.189\,\mathrm{T}^{-2}$
Coil current per coil giving $\lvert\Delta f/f\rvert$ = mean beat offset: γ_L; $\gamma = 10^{-22}$	6.5×10^{16} A; 1.0×10^{14} A

Table 9: Higher-current cases (per coil). Source: `csv/fixed_case_and_feasibility.csv`. Not operating recommendations.

I (A)	$U = IR$ (V)	P_8 (kW)	Onderdonk t (s)	I/I_{cont}	B_{main} (mT)	$\Delta f/f$ (γ_L)
1.5	2684	32.2	11.7	3.7	1.25	-1.38×10^{-44}
2	3578	57.2	6.6	4.9	1.66	-2.45×10^{-44}
3	5367	128.8	2.9	7.4	2.49	-5.52×10^{-44}

5 Discussion

Status of the prediction. Subject to the numerical limitations and control qualifications listed below, the calculated pure even field-energy signal of $\approx -1.20 \times 10^{-48}$ is numerically supported; it is a usable conditional design prediction, not a validated measurement result [11]. It is an acceptable calculated order of magnitude within the chosen ideal current-sheet model, not a TAL detection and not a measured instrument sensitivity. On the comparison scales of Table 8, the signal could not be measured with the OCXO pair, nor with a Sr reference clock.

Limitations.

1. *No measurement data.* No ON/OFF/$\pm I$ phase or frequency time series, state log, or switching-correlated thermal measurement exists yet; the dimensional measurement record (Appendix A) does not replace these. The actual sensitivity of the apparatus is therefore undetermined. The next required evidence is a timestamped phase plus state-label ON/OFF estimate with its uncertainty, not another calculated signal-to-beat ratio.

2. *Mean beat offset.* The 65-min Lissajous cycle gives a mean beat offset of 2.56410×10^{-11} only if it is a true 2π electrical phase cycle. It is a mean beat offset, not a noise floor. Tying the switching schedule to the beat period is not intrinsically invalid, but the offset does not

Table 10: Median and mean $\chi^2_{\rm red}$ over 123 galaxies (2596 radii) for the six baryonic baselines and the TAL factor (seventh model). Sources: `hetedik_modell_kimenet/comparison.json` (`stats`, `variant_stats`), `hetedik_modell_kimenet/run_meta.json`, `tal_factor_7th_model.md` §4.

Model	$V_{\rm pred}$	Median	Mean
vbar	$V_{\rm bar}$	3.1433	4.2161
vbar_095	$0.95\,V_{\rm bar}$	3.1995	4.3725
vbar_105	$1.05\,V_{\rm bar}$	2.9555	4.3224
no_bulge	$(V_{\rm gas}^2+V_{\rm disk}^2)^{1/2}$	3.3131	6.3928
no_gas	$(V_{\rm disk}^2+V_{\rm bul}^2)^{1/2}$	4.0474	5.2533
disk_only	$\lvert V_{\rm disk}\rvert$	4.2623	7.5707
TAL, $L_0 = 1.2\times10^{-11}$ m (primary)	$f_L^{1/2}V_{\rm bar}$	3.1433	4.2161
TAL, $L = 367$ m / $\gamma = 10^{-22}$	forward	3.1433	4.2161
TAL, $L = 2.36$ kpc	forward	3.1433	4.2161
TAL, fitted $L = 2.371$ Mpc (in-sample, not blind)	forward	2.7083	4.1761
TAL, fitted $L = 0.473$ Mpc (in-sample, not blind)	inverse	3.1425	4.3265

make 65 minutes optimal; the interval must be justified through noise and thermal-settling tests.

3. *Far-field convergence.* The four runs use identical nlo and nt settings in this region. The zero sample standard deviation is therefore not a far-field error estimate; far-field convergence must be tested by varying nlo and nt separately. A joint refinement ($n_{\rm lo}$ 6000→12 000, n_t 32→64) changes ΔW by -9.1×10^{-7} to -3.0×10^{-4} relative at $L = 2.6$–100 m (Appendix B); separate variation has not yet been performed.

4. *Matching residual at 0.95 m.* The residual of $0.00141340\,\mathrm{J\,A^{-2}}$ (0.0649 % of the inner value) remains a numerical limitation; the curve is not continuous to machine precision.

5. *Background cross term.* The pure coil-B^2 term is even under polarity reversal. In a constant background field, the change in energy density also contains a $2B_{\rm bg}\cdot B_{\rm coil}$ cross term; ideally, averaging equal-magnitude $\pm I$ states relative to OFF cancels this term. The reported -1.20×10^{-48} value refers to this isolated even term, not to the full prediction for every possible simple ON/OFF measurement.

6. *Thermal control.* Joule heating is also even in the current, so polarity reversal does not separate it. The proposed control dissipates the same heat load (2.81 W) in a non-field-generating (bifilar) resistor near the coils during the OFF intervals. The purpose of the equal-total-power control with a small residual magnetic field is to reduce switching-correlated thermal changes. Temperature matching between the ON and OFF states and the residual field of the control must be verified by measurement; any remaining mismatch must be included in the uncertainty of the frequency estimate. Equal total power does not guarantee identical spatial heat deposition or identical conductive and radiative coupling to the clocks.

7. *Model idealisation.* The field is that of ideal current sheets with the measured winding diameter; the actual multilayer winding, current tolerances, magnetic materials, and the environment are not modelled, and none of the quoted numerical standard errors includes

these effects. The winding diameter rests on an author-supplied measurement record that was not independently certified on site.

8. *Safety.* The Onderdonk estimate is only an idealized adiabatic fusing time; no safe pulse duration follows from it, and the calculated high-current cases of Table 9 are not operating recommendations.

9. *Theory inputs.* The bound $\gamma < 10^{-22}\,\mathrm{m^3\,kg^{-1}}$ is the source's own estimate (Eq. (3) of [9]). How f_L enters galactic rotation curves is not specified in the published material; the SPARC comparison rests on stated implementation choices and on a 123-galaxy subsample, not on all 175 galaxies.

Measurement sequence implied by the design. The design calls for (1) a 12 h baseline without coil current, logging the Lissajous phase of the two OCXOs, battery voltage and current, and housing temperature; (2) equal-length ON/OFF segments at 14 mA for at least 12 h; (3) the same cycle with reversed polarity, so that effects linear in B change sign while B^2 effects do not; (4) evaluation of ON$-$OFF differences and their uncertainties and of polarity sums and differences, with separate treatment of temperature and supply couplings; and (5) the matched-heat-load control above, with only post-settling segments entering the evaluation (`report_hu.md` §4). Within this sequence the TAL prediction for the isolated even term is -1.202×10^{-48}.

6 Conclusion

For a fixed, fully specified laboratory configuration, the TAL relations $f_L = \exp(-\gamma_L \bar{\rho}_L)$, $\gamma_L = GL^2/c^2$, with the magnetic term $B^2/(2\mu_0 c^2)$, predict a differential fractional frequency shift of the isolated, polarity-even coil-B^2 term of $\approx -1.20 \times 10^{-48}$ between a main and a reference oscillator at 14 mA per coil. This value is reproduced by an independent Sobol cubature to $-0.0722\,\%$, and the previous outer-integral discrepancy near $L \approx 3$ m has been eliminated at the checked points. The remaining numerical limitations (untested far-field resolution dependence at fixed nlo/nt, and the 0.0649 % matching residual at 0.95 m) are disclosed. The result is a conditional design prediction in an ideal current-sheet model, not a measured sensitivity and not a detection; the next required evidence is an ON/OFF estimate with uncertainty from timestamped phase data and state labels. In the repository's SPARC analysis, the TAL factor at the published scales is numerically identical to the pure-baryon baseline.

Data availability

All inputs, outputs and review texts are archived in Zenodo record V20, "Time Attraction Law V20 — audited release with independent third-round review", doi:10.5281/zenodo.23540741 [11]; the previous version V19 (seventh-model SPARC analysis) is doi:10.5281/zenodo.23497128 [12]. The GitHub V18 package, identical in code and CSV/JSON files to the V20 archive and additionally containing the two text corrections of the third-round review, is available at https://github.com/unbekanntedavid-glitch/Time-Attraction, commit `042a616596f1204e78ef137da10c262761bb922e` (directory `V18/`) [13]. SHA-256 checksums are provided in `V18/SHA256SUMS_V18.txt`, `V18/SHA256SUMS_ZENODO_`

V20.txt and V18/final/SHA256SUMS.txt. SPARC rotation curves are available from the SPARC database [14]. All materials are licensed under CC BY-NC 4.0. No clock phase time series exist yet.

Code availability

The calculation is reproduced by code/final_fixed.py with the modules code/fields.py, code/energy_integrator.py, and code/tophat_finite_source.py, written in Python 3 with NumPy [23] and SciPy [17], all in V18/final/. The computational checks of Appendix B are regenerated by V18/make_revision3_checks.py. The SPARC seventh-model scripts and outputs are referenced in tal_factor_7th_model.md [12, 13]. Figures 1 and 2 were regenerated from the archived CSV files with a plotting script supplied with this manuscript source.

Declarations

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Competing interests. The author has no competing interests to declare.
Author contributions. D. Sz. is the sole author: theory, apparatus specification, measurements of the apparatus dimensions, calculations and writing.
Ethics approval, consent. Not applicable.
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A Measurement record: winding diameter

Table 11: Winding-diameter measurement record (author-supplied; report_hu.md §1.2).

Item	Value
Date of measurement	2025-11-13
Measured by	D. Szabolcsi
Instrument (winding diameter)	caliper, resolution 0.02 mm
Signal measurement	PicoScope 2205A and Rigol DS1202ZE oscilloscopes
Readings (5 consecutive)	291.7; 291.7; 291.7; 291.7; 291.7 mm
Mean	291.7 mm
Sample standard deviation	0 mm (all five readings identical)
Location of the 300 mm dimension	outer diameter of the coil-body flange, not the winding surface

The zero sample standard deviation of five identical readings does not imply zero total measurement uncertainty of the dimension. Consequences for the geometry: the coil-body inner edge stays at 0.30 m and the axis circle at $0.30 + 0.15 = 0.45$ m; the reference is 1 m from the outer flange surface (0.60 m), i.e. 1.60 m from the main clock and 1.004 m from the winding surface; the gap between granite and flange is 8 cm (8.42 cm to the winding surface); the gaps between

adjacent coil bodies and windings are 4.4 cm and 5.3 cm. The resistance computed from the measured diameter, 1788.160 Ω, differs from the stated 1789 Ω by −0.0470 % [11].

B Summary of the revision-3 computational checks

`V18/revision3_checks.json` was *regenerated by the author's pipeline* on 2026-10-21 by running `V18/make_revision3_checks.py` on the `V18/final` package. It is not the reviewer's original file; the reviewer's script `check_revision3.py` is not available. Its contents are computational checks only, with no raw clock data (Table 12).

Table 12: Contents of `V18/revision3_checks.json` (author-regenerated).

Check	Result
SHA-256 integrity	18/18 entries match `V18/final/SHA256SUMS.txt`
B, main clock	discrete loops $1.1632900301 \times 10^{-5}$ T vs. package $1.1632899803 \times 10^{-5}$ T; rel. diff. -4.28×10^{-8}
B, reference clock	$4.3053923117 \times 10^{-7}$ T vs. $4.3053922792 \times 10^{-7}$ T; rel. diff. -7.56×10^{-9}
Main-peak rerun, $L = 0.6309573$ m ($n_{\mathrm{hi}} = 24\,000$)	$\Delta W/I^2 = 1.9611445\,\mathrm{J\,A^{-2}}$; $k = -6.131177672606294 \times 10^{-45}\,\mathrm{A^{-2}}$ (identical to JSON); signal $-1.2017108 \times 10^{-48}$; runtime 56.9 s
Outer integral, n_{lo} 6000→12 000 and n_t 32→64 jointly	rel. change -9.08×10^{-7} (2.6 m), -1.95×10^{-6} (3.0 m), -2.71×10^{-5} (10 m), -3.02×10^{-4} (100 m)
Route A vs. B at 3.0 m	rel. diff. -7.44×10^{-16}
Residual at $L = 0.95$ m	$0.0014133960\,\mathrm{J\,A^{-2}}$ (0.06491 % of inner value); equals $W_{\mathrm{main}} - W_{\mathrm{ref}}$ total-energy difference to 8.9×10^{-16}

The joint refinement in the outer-integral row changes nlo and nt together; it does not replace the separate variation of nlo and nt requested by the review.

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